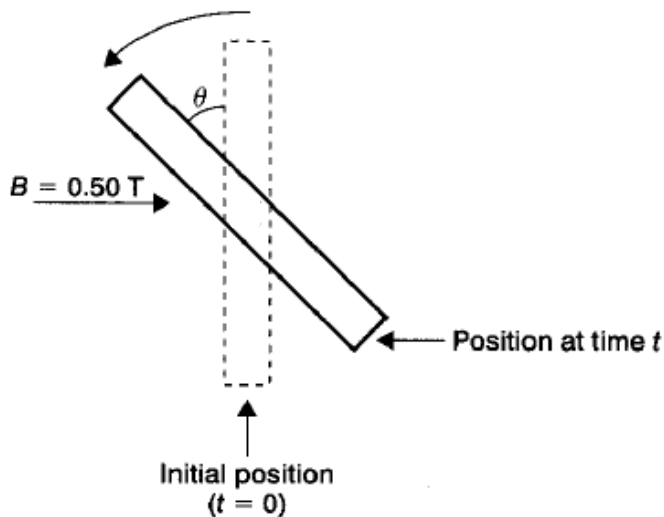


- 24** A flat, circular coil of wire 30 turns, each of area 0.025 m^2 , is initially placed with its plane at right angles to a uniform magnetic field of flux density 0.50 T .



What is the magnitude of the induced e.m.f. in the coil when $t = 1.0 \text{ s}$, if the coil is rotating at 60° s^{-1} ?

	A	0.20 V	B	0.34 V
	C	0.38 V	D	0.39 V

Ans: B

$$\varphi = NBA = NBA \cos \omega t$$

$$\omega = \frac{60}{180} \times \pi = 1.05 \text{ rad s}^{-1}$$

$$|\varepsilon| = N \frac{\Delta \varphi}{\Delta t} = NBA \omega \sin \omega t = 30(0.50)(0.025)(1.05) \sin 60(1) = 0.34 \text{ V}$$

$$\text{Option A: } NBA \omega \cos \omega t = 30(0.50)(0.025)(1.05) \cos 60(1) = 0.20 \text{ V}$$

$$\text{Option C: } \varphi = NBA = 0.38 \text{ V}$$

$$\text{Option D: } NBA \omega = 30(0.50)(0.025)(1.05) = 0.39 \text{ V}$$